

DSAT Practice — Detail (According to the Text)

25 Original MCQs | Questions Only

Q1.

[Easy]

The Kepler space telescope, launched by NASA in 2009, was designed to detect exoplanets by monitoring the brightness of distant stars. When a planet passes in front of its host star, it causes a small, periodic dip in the star's brightness—a phenomenon known as a transit. Over the course of its mission, Kepler identified more than 2,600 confirmed exoplanets, fundamentally transforming scientists' understanding of how common planets are in the Milky Way galaxy.

According to the text, how does the Kepler space telescope detect exoplanets?

- A) By photographing planets directly using high-resolution imaging technology.
 - B) By analyzing the chemical composition of planetary atmospheres using spectroscopy.
 - C) By detecting gravitational waves produced when planets collide with asteroids.
 - D) By measuring periodic decreases in a star's brightness caused by a planet crossing in front of it.
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Q2.

[Easy]

In many traditional Japanese gardens, carefully raked gravel represents flowing water, and strategically placed rocks symbolize mountains or islands. These gardens are not intended to replicate natural landscapes literally; rather, they are designed to evoke the essence of nature through abstraction. Visitors are encouraged to view the garden from a fixed vantage point, such as a wooden veranda, rather than walking through it.

According to the text, what is the intended purpose of the gravel and rocks in traditional Japanese gardens?

- A) To abstractly suggest elements of the natural world, such as water and mountains.
 - B) To provide a durable surface that requires no maintenance.
 - C) To recreate a specific natural landscape with precise geographic accuracy.
 - D) To mark paths that guide visitors through the garden on foot.
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Q3.*[Easy]*

The axolotl, a salamander native to lakes near Mexico City, possesses a remarkable ability to regenerate lost body parts, including limbs, portions of its heart, and even sections of its spinal cord. Unlike most amphibians, the axolotl retains its larval features throughout its life—a condition known as neoteny—and remains fully aquatic. Because of its regenerative capabilities, the axolotl has become a valuable subject in biomedical research.

According to the text, what is neoteny?

- A) The ability to regenerate lost limbs and organs.
 - B) The retention of larval features into adulthood.
 - C) The process by which amphibians transition from aquatic to terrestrial habitats.
 - D) A condition in which an organism grows larger than typical members of its species.
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Q4.*[Easy]*

Sociologist Anna Kessel studied the effects of four-day workweek trials in several Scandinavian companies. She found that employees who worked four days per week reported higher job satisfaction and lower burnout than those who worked five days. Importantly, the companies participating in the trial did not experience a decrease in overall productivity, as employees tended to work more efficiently during their reduced hours.

According to the text, what did Kessel find about the participating companies' productivity?

- A) Productivity varied significantly depending on the industry in which each company operated.
 - B) Productivity increased dramatically, prompting all participating companies to adopt the four-day model permanently.
 - C) Productivity declined initially but recovered after six months of adjustment.
 - D) Productivity did not decline, because employees worked more efficiently during the shorter workweek.
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Q5.*[Easy]*

The Library of Ashurbanipal, assembled in the seventh century BCE in the Assyrian capital of Nineveh, is one of the oldest known collections of organized texts. The library contained approximately 30,000 clay tablets inscribed with cuneiform script, covering subjects ranging from astronomical observations to medical treatments to mythological narratives, including a version of the Epic of Gilgamesh.

According to the text, what materials were the texts in the Library of Ashurbanipal written on?

- A) Papyrus scrolls.
 - B) Clay tablets.
 - C) Silk fabric.
 - D) Animal hide parchment.
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Q6.*[Easy]*

Meteorologist Hana Otsuka explains that lake-effect snow occurs when cold air masses move across large bodies of warm water. As the air absorbs moisture and heat from the lake surface, it becomes unstable, rises, and forms dense snow clouds that deposit heavy snowfall on the lake's downwind shore. Cities on the eastern shores of the Great Lakes, such as Buffalo, New York, routinely receive some of the highest annual snowfall totals in the United States as a result of this phenomenon.

According to the text, which factor directly contributes to the formation of lake-effect snow?

- A) Extended periods of below-freezing temperatures that cause lake surfaces to freeze entirely.
 - B) Warm air masses being deflected by mountain ranges near the Great Lakes.
 - C) Ocean currents carrying tropical moisture to the Great Lakes region.
 - D) Cold air masses absorbing moisture and heat as they pass over warm lake surfaces.
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Q7.*[Easy]*

The Rosetta Stone, discovered in Egypt in 1799, bears the same decree inscribed in three scripts: hieroglyphic, Demotic, and ancient Greek. Because scholars could already read ancient Greek, they used the Greek text as a key to decipher the hieroglyphic script. French linguist Jean-François Champollion is credited with achieving the breakthrough decipherment in 1822.

According to the text, why was the Greek text on the Rosetta Stone important for decipherment?

- A) Because scholars already understood Greek and could use it to decode the hieroglyphic script.
 - B) Because Greek was the oldest of the three scripts on the stone.
 - C) Because the Greek text contained additional information not found in the other two scripts.
 - D) Because Champollion was a native Greek speaker who recognized the text immediately.
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Q8.*[Easy]*

Behavioral ecologist Tomoko Igarashi studied alarm calls in a population of vervet monkeys in East Africa. Igarashi observed that vervets produce acoustically distinct alarm calls for different predator types: one call for eagles, a different call for leopards, and a third for snakes. Each call triggers a specific behavioral response—for example, the eagle alarm causes monkeys to look up or hide in low bushes, while the leopard alarm causes them to climb trees.

According to the text, how do vervet monkeys respond to the eagle alarm call?

- A) They remain motionless on the ground until the threat passes.
 - B) They climb to the tops of nearby trees.
 - C) They look upward or seek cover in low vegetation.
 - D) They produce a series of additional alarm calls to warn neighboring groups.
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Q9.

[Medium]

Economist Priya Nair has argued that microfinance—the provision of small loans to low-income entrepreneurs in developing countries—does not consistently deliver the transformative economic outcomes that its proponents have claimed. Reviewing data from randomized controlled trials in six countries, Nair found that while microloan recipients did tend to invest in small business activities, their overall household income and consumption levels did not increase significantly compared to control groups. Nair suggests that the loans may be too small to generate the economies of scale needed for sustained business growth.

What does the text most strongly suggest about the effect of microloans on recipients' household income?

- A) Microloans increased household income in the short term but caused financial losses over longer periods.
 - B) Microloans substantially increased household income, but only in countries with strong regulatory frameworks.
 - C) Microloans did not lead to a meaningful increase in household income relative to those who did not receive loans.
 - D) Microloans had no measurable effect on any aspect of recipients' economic behavior.
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Q10.

[Medium]

Archaeologist Liam Gallagher has proposed that the alignment of Neolithic stone circles in northern Scotland was dictated not by astronomical phenomena, as many scholars have assumed, but by the topography of the surrounding landscape. Gallagher notes that the gaps between standing stones at several sites correspond to prominent natural features on the horizon—mountain passes, river valleys, and distinctive rock formations—rather than to the positions of celestial bodies at solstices or equinoxes.

Based on the text, how does Gallagher's interpretation of stone circle alignments differ from the prevailing view?

- A) Gallagher contends the circles were aligned randomly, with no deliberate orientation of any kind.
 - B) Gallagher argues the circles were built for defensive purposes rather than ceremonial ones.
 - C) Gallagher believes the circles were constructed more recently than other scholars have proposed.
 - D) Gallagher attributes the alignments to landscape features rather than to the positions of the sun and stars.
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Q11.

[Medium]

Marine biologist Lena Kowalski studied the feeding behavior of manta rays in the Coral Triangle and discovered that these animals do not filter-feed indiscriminately, as was previously believed. Instead, Kowalski's tracking data showed that manta rays actively seek out areas of high zooplankton density, adjusting their swimming depth and trajectory to maximize food intake. She also found that mantas returned to the same productive feeding sites at consistent times of day, suggesting they possess some form of spatial and temporal memory.

What does the text most strongly suggest about manta rays' feeding behavior?

- A) Manta rays feed exclusively at night, when zooplankton concentrations are highest.
 - B) Manta rays compete with other species for access to high-density zooplankton patches.
 - C) Manta rays deliberately target zooplankton-rich areas rather than feeding passively on whatever they encounter.
 - D) Manta rays consume a wider variety of organisms than zooplankton.
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Q12.

[Medium]

Political scientist Mei Tanaka has documented what she calls 'policy feedback loops' in public healthcare systems. Once a universal healthcare policy is established, Tanaka argues, it creates a large constituency of beneficiaries—patients, healthcare workers, and administrators—who have a vested interest in maintaining and expanding the program. This constituency then exerts political pressure that makes the policy increasingly difficult to repeal, even when economic conditions change or political leadership shifts.

According to the text, what makes universal healthcare policies difficult to repeal, in Tanaka's view?

- A) The large group of beneficiaries created by the policy exerts political pressure to preserve it.
 - B) International treaties require participating nations to maintain healthcare programs indefinitely.
 - C) The administrative costs of dismantling the program exceed the savings from repealing it.
 - D) Voters generally lack sufficient information about healthcare policy to support repeal efforts.
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Q13.

[Medium]

Architect Rosa Martín designed the Canopy Pavilion, a public gathering space in Valencia, Spain, whose roof is composed of interlocking ceramic tiles arranged in a pattern inspired by the scales of a pinecone. The tiles are angled so that rainwater is channeled into a central collection basin for use in irrigating the surrounding gardens. Martín has stated that the pavilion's design reflects her conviction that architecture should serve both aesthetic and ecological functions simultaneously, rather than treating sustainability as an afterthought.

According to the text, what is the functional purpose of the angled ceramic tiles on the Canopy Pavilion's roof?

- A) To insulate the interior of the pavilion against extreme summer temperatures.
 - B) To generate electricity by capturing solar energy as it strikes the roof surface.
 - C) To direct rainwater into a basin where it is collected for garden irrigation.
 - D) To reduce wind resistance during seasonal storms common to the Valencia region.
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Q14.

[Medium]

Cognitive psychologist Elena Rossi conducted an experiment in which participants were asked to solve anagram puzzles under two conditions: one group worked in silence, and the other listened to instrumental music at a moderate volume. Rossi found that participants in the music condition solved anagrams significantly faster than those in the silence condition, but only when the anagrams were of moderate difficulty. For the most challenging anagrams, performance was equivalent across both groups, leading Rossi to conclude that background music enhances cognitive performance up to a point, beyond which task difficulty overwhelms any benefit the music provides.

What does the text indicate about the effect of background music on the most challenging anagrams?

- A) Music did not improve performance on the most difficult puzzles compared to working in silence.
 - B) Music improved performance on difficult puzzles even more than on moderate ones.
 - C) Music significantly impaired participants' ability to solve difficult puzzles.
 - D) Music distracted participants from the most difficult puzzles, causing them to abandon the task.
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Q15.

[Medium]

Historian Youssef El-Amine has examined the role of coffeehouses in seventeenth-century Ottoman intellectual life. Contrary to the widespread characterization of coffeehouses as venues primarily for leisure and socializing, El-Amine presents evidence that many coffeehouses functioned as informal academies where poets, scholars, and administrators debated philosophy, exchanged manuscripts, and discussed matters of state. He argues that the coffeehouse was a critical but underappreciated institution in the Ottoman intellectual ecosystem.

Based on the text, what is El-Amine's main argument about Ottoman coffeehouses?

- A) They were primarily used by government officials to conduct state business.
 - B) They served as important intellectual centers, not merely as social gathering places.
 - C) They were more common in the Ottoman Empire than in European countries during the same period.
 - D) They declined in cultural importance during the eighteenth century.
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Q16.

[Medium]

Entomologist Clara Voss discovered that a species of parasitoid wasp in Costa Rica lays its eggs exclusively inside the larvae of a particular beetle species. When the wasp larvae hatch, they consume the beetle larva from the inside. Remarkably, the wasp larvae initially avoid consuming the beetle's vital organs, keeping the host alive—and therefore fresh—until the wasp larvae are nearly ready to pupate. Only at the final stage of development do the wasp larvae consume the remaining tissue, killing the host.

According to the text, why do the wasp larvae initially avoid consuming the beetle larva's vital organs?

- A) The vital organs contain toxins that are harmful to the wasp larvae during early development.
 - B) Keeping the host alive preserves the freshness of the tissue the wasp larvae will consume later.
 - C) The beetle larva's immune system protects its vital organs from being consumed.
 - D) The wasp larvae are too small during early development to consume large organs.
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Q17.

[Medium]

Psycholinguist Omar Said investigated whether the grammatical gender of nouns influences how speakers of gendered languages perceive objects. In one experiment, Spanish speakers and German speakers were asked to describe a key (which is masculine in German—*der Schlüssel*—and feminine in Spanish—*la llave*). Spanish speakers tended to use adjectives like 'delicate' and 'elegant,' while German speakers described keys as 'sturdy' and 'heavy.' Said interpreted these results as evidence that grammatical gender can subtly shape the associations speakers form with everyday objects.

According to the text, what pattern did Said observe in the experiment?

- A) Speakers' adjective choices for the same object varied in ways that corresponded to the object's grammatical gender in their language.
 - B) Spanish speakers and German speakers used identical adjectives but applied them to different objects.
 - C) Speakers of both languages described keys using adjectives associated with the masculine grammatical gender.
 - D) German speakers were more likely than Spanish speakers to describe objects using abstract rather than concrete adjectives.
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Q18.

[Medium]

Engineer Fatima al-Rashid designed a portable water purification device powered entirely by solar energy. The device uses ultraviolet light to neutralize waterborne pathogens without requiring chemical additives or replacement filters. In field trials conducted in rural communities in Senegal, the device purified an average of 20 liters of water per hour—enough to meet the daily drinking water needs of a family of four. Al-Rashid notes, however, that the device's output drops significantly on overcast days, limiting its reliability in regions with extended cloudy seasons.

According to the text, what limitation does the purification device have?

- A) Its water output decreases substantially when sunlight is reduced by cloud cover.
 - B) It requires chemical additives to purify water containing certain types of pathogens.
 - C) It cannot purify enough water to meet the daily needs of more than one person.
 - D) Its filters must be replaced after every 100 liters of purified water.
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Q19.*[Hard]*

Neuroscientist Arjun Mehta used functional magnetic resonance imaging (fMRI) to study brain activity in professional chess players and novices while both groups evaluated chess positions. Mehta found that professionals showed heightened activation in the fusiform face area (FFA)—a brain region typically associated with facial recognition—when viewing chess configurations. Novices, by contrast, showed no significant FFA activation. Mehta hypothesizes that through years of practice, experts develop holistic pattern recognition for chess positions analogous to the rapid, automatic process by which most people recognize faces.

What does the text most strongly imply about the FFA's role in expert chess players' cognition?

- A) It enables chess players to remember the faces of opponents they have previously competed against.
 - B) It has been repurposed through extensive practice to support rapid pattern recognition of chess configurations.
 - C) It is activated equally in experts and novices but processes information at different speeds.
 - D) It functions identically in chess cognition and in facial recognition, with no specialization for either task.
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Q20.*[Hard]*

Literary critic Nkechi Okafor has argued that the narrator of Chinua Achebe's *Things Fall Apart* employs a deliberately restrained style—avoiding elaborate metaphors and subjective commentary—in order to mirror the communal, oral storytelling traditions of the Igbo people depicted in the novel. Okafor contends that this restraint is itself a form of cultural fidelity: by allowing events to unfold without heavy authorial interpretation, the narrative voice reproduces the communal narrator's expectation that listeners will draw their own moral conclusions from the story.

Based on the text, what does Okafor suggest is the purpose of the narrator's restrained style?

- A) To allow the novel to be easily translated into multiple languages without losing stylistic nuance.
 - B) To compensate for Achebe's limited familiarity with Western literary conventions at the time of writing.
 - C) To reflect Igbo oral traditions in which communal narrators present events without imposing explicit moral judgments.
 - D) To signal the narrator's disapproval of the cultural practices depicted in the novel.
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Q21.*[Hard]*

Geochemist Hiro Watanabe has proposed that the unusually high concentration of iridium found in a thin clay layer at the Cretaceous-Paleogene (K-Pg) boundary—the geological marker of the mass extinction event 66 million years ago—may not have originated exclusively from the asteroid impact at Chicxulub, as the standard model asserts. Watanabe points to evidence that massive volcanic eruptions in the Deccan Traps of present-day India, which occurred during the same geological period, also release iridium. He argues that the K-Pg iridium layer may therefore represent a combined signature of both events rather than an exclusively asteroidal one.

What does the text indicate about Watanabe's position on the iridium at the K-Pg boundary?

- A) He disputes that an asteroid impact occurred at Chicxulub 66 million years ago.
 - B) He argues that volcanic activity, not just the Chicxulub asteroid impact, may have contributed iridium to the boundary layer.
 - C) He claims that the Deccan Traps eruptions alone account for the iridium concentration at the K-Pg boundary.
 - D) He argues that the K-Pg boundary layer contains no iridium and has been misidentified.
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Q22.*[Hard]*

Economist Adaora Ibe studied the long-term effects of colonial-era railroad construction in sub-Saharan Africa. Ibe found that districts located within 20 kilometers of a colonial railroad line had, on average, 15 percent higher per capita income in 2015 than otherwise similar districts farther from the rail network. However, Ibe also found that these same near-railroad districts exhibited greater income inequality: the benefits of proximity to transportation infrastructure were concentrated among landowners and merchants, while subsistence farmers saw little improvement in their economic circumstances.

Based on the text, what is a key nuance in Ibe's findings about colonial railroads?

- A) The economic effects of railroads were uniformly positive and affected all social groups equally.
 - B) Railroads benefited only subsistence farmers, leaving landowners and merchants unaffected.
 - C) Districts near railroads had lower income inequality than districts farther from the rail network.
 - D) While proximity to railroads increased average income, the gains were not distributed equally across social groups.
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Q23.

[Hard]

In a 2023 study, psychologist Bianca Torres examined whether exposure to nature documentaries affects viewers' environmental attitudes. Torres randomly assigned participants to watch either a nature documentary or a control video about interior design. After viewing, participants completed a survey measuring their willingness to pay higher taxes for environmental conservation. Torres found that documentary viewers expressed significantly greater willingness to pay than control-group viewers, but only among participants who had reported low prior engagement with environmental issues. Participants who were already highly engaged showed no change, suggesting a ceiling effect.

What does the text indicate about participants who were already highly engaged with environmental issues before the study?

- A) They showed the largest increase in willingness to pay of any group in the study.
 - B) They were excluded from the study because their existing attitudes would have biased the results.
 - C) They expressed less willingness to pay higher taxes after watching the documentary than before.
 - D) Watching the nature documentary did not increase their willingness to pay higher conservation taxes.
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Q24.

[Hard]

Biochemist Soren Lindgren investigated the mechanism by which certain deep-sea bacteria survive in environments with extremely high hydrostatic pressure. Lindgren discovered that these bacteria produce a unique class of membrane lipids—piezolipids—whose molecular structure remains flexible under pressures that would rigidify the membranes of surface-dwelling bacteria, causing cell death. When Lindgren genetically modified a surface-dwelling species to produce piezolipids, the modified bacteria survived pressures up to 400 atmospheres, whereas unmodified bacteria of the same species perished above 100 atmospheres.

Based on the text, what role do piezolipids play in the survival of deep-sea bacteria?

- A) They generate metabolic energy that allows the bacteria to move through dense water columns.
 - B) They maintain membrane flexibility under extreme pressure, preventing the cellular rigidity that would otherwise be fatal.
 - C) They protect the bacteria's genetic material from mutations caused by high-pressure environments.
 - D) They enable deep-sea bacteria to photosynthesize in the absence of sunlight.
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Q25.

[Hard]

Art historian Rina Chakraborty has reexamined a collection of bronze sculptures from the Chola dynasty (9th–13th century CE) in southern India, focusing on the patinas—surface coloration caused by oxidation over centuries. Previous scholars treated the green and brown patinas as incidental effects of aging, but Chakraborty argues that Chola artisans deliberately controlled the alloy composition of each sculpture to produce specific patina colors over time. She supports this claim by demonstrating that sculptures depicting benevolent deities consistently use copper-rich alloys (which produce green patinas), while sculptures of fierce deities use tin-rich alloys (which produce darker, brown patinas)—a pattern too systematic to be coincidental.

According to the text, what evidence does Chakraborty cite to support her argument that Chola artisans intentionally controlled patina colors?

- A) Chemical analysis showing that all Chola sculptures used identical alloy compositions regardless of subject matter.
 - B) Written records from Chola workshops describing techniques for accelerating oxidation on bronze surfaces.
 - C) A consistent correspondence between alloy composition and the type of deity depicted, suggesting the color outcomes were planned.
 - D) Comparison with contemporary Indian sculptors who use the same alloy-selection techniques today.
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